Math 651 Spring 2021

Take-home Final Exam

Due: Wednesday, April 28, 12:00 noon

In doing this exam, you may refer freely to your Math 651 class notes, the course textbooks, and any of the classes previous homework for this course, including any comments and notes on them. You may not use any other references, including web-based, printed, or human ones. The one exception to this rule is that you may refer to non-human TEX resources that might help you present your work better.

You are not to discuss your work with anyone other than your instructor. In particular you should not discuss the problems with any students, whether they are in this class or not, or any other professors. If you have any doubts as to what is allowed, you are responsible for asking for clarification.

Solutions should be presented with a good mathematical writing style, and must be typeset. Use a proposition/proof format whenever appropriate. If you need to use a result in the textbooks, be sure sure you explicitly refer to it, giving both its number and page number. Clarity of presentation will be a factor in grading.

The problems are not in order of difficulty, and you may not get them all.

1. (15 pts.) Let $S = \{X_\alpha\}_{\alpha \in \mathcal{A}}$ be a collection of non-empty topological spaces.

   Theorem 1 (Munkres, p. 205)**.** *If* $\prod_{\alpha \in \mathcal{A}} X_\alpha$ *is Hausdorff, regular, or normal, then* X_α *has the same property.*

   Proof. Let $P = \prod_{\alpha \in A} X_\alpha$ be the product of a collection of non-empty topological spaces; let $p : P \to X_\beta$ be projection onto a particular factor. Suppose there are distinct points $x, y \in X_\beta$ and closed sets E, F such that $x \in E$ and $y \in F$. Then, there are distinct points $\vec{x}, \vec{y} \in P$ such that $p(\vec{x}) = x$ and $p(\vec{y}) = y$.

   Suppose P is Hausdorff. Then, there exist disjoint neighborhoods U of $\vec{x}$ and V of $\vec{y}$ in P. Since p is an open map, it follows that $p(U)$ and $p(V)$ are disjoint neighborhoods of x and y in X_β.

   Suppose P is regular, and suppose F does not contain x. Then, $p^{-1}(F)$ is closed set in P which does not contain $\vec{x}$. By regularity, there exist disjoint open sets U containing x and V containing $p^{-1}(F)$. Since p is an open map, it follows that $p(U)$ and $P(V)$ are open sets containing x and F.

   Suppose P is normal, and suppose E does not intersect F. Then, $p^{-1}(E)$ and $p^{-1}(F)$ are disjoint closed sets containing $\vec{x}$ and $\vec{y}$, respectively. By normalcy, there are disjoint open sets U and V containing $p^{-1}(E)$ and $p^{-1}(F)$, respectively. Since p is an open map, it follows that $p(U)$ and $p(V)$ are open sets containing E and F, respectively. □

2. (15 pts.) Let $p : E \to B$ be a covering map.

Theorem 2. *The following are true:*

(a) p is open.

(b) If B is Hausdorff, then E is Hausdorff.

(c) E is compact if, and only if, B is compact and all fibers of p are finite.
(You may want to first assume B is connected. Establishing the claim under that assumption implies it more generally by applying it to each of the finitely-many connected components of B and their inverse images in E. But you'll need to explain why there are only finitely-many connected components of B.)

(d) If E is Hausdorff and compact, then B is Hausdorff.
(Note that the assumption that E is compact cannot be dropped. Example: Let $E = \mathbb{R}^2 \smallsetminus \{\mathbf{0}\}$, and let G be the group generated by $(x, y) \mapsto (2x, y/2)$. This acts properly discontinuously, and $p : X \to B = E/G$ is a covering map, but B is not Hausdorff — neighborhoods of images of points on the $x-$ and $y-$ axes intersect.)

Proof. Let $p : E \to B$ be a covering map.

(a) Let U be an open set in E, and consider a point b in its image $p(U) \subseteq B$ arising from some $x \in U$ such that $p(x) = b$.
By the covering property, there is an open neighborhood V of $p(x)$ that is evenly covered via p by $p^{-1}(U) = \bigcup_{\alpha\in}\{V_\alpha\}_{\alpha\in E}$ which cover E such that $p|_{V_\alpha} : V_\alpha \to V$ is a homeomorphism.
The point x is contained in one of these open neighborhoods V_a and thus in some open neighborhood $V_\alpha \cap U \subseteq V_\alpha$ whose image is an open neighborhood $p(V_\alpha \cap U) \subseteq p(U)$ of $p(x)$. Since the set $p(U)$ can thus written as a union of open sets, it follows that $p(U)$ is itself open.

(b) Suppose B is Hausdorff, and consider distinct points x and y in E.
Suppose $p(x) = p(y) = b$ and let U be a neighborhood of b that is evenly covered via p by $p^{-1}(U) = \sqcup_{\alpha\in A}U_\alpha$ for disjoint neighborhoods U_α which are homeomorphic to U under p. In particular, x and y are contained in disjoint neighborhoods U_α.
Otherwise, suppose $p(x) \neq p(y)$. By the Hausdorff property, there are disjoint open sets U of $p(x)$ and V of $p(y)$. Since p is a covering map, there are open neighborhoods V_1 and V_2 of $p(x)$ and $p(y)$ which are evenly covered by p. They yield disjoint open neighborhoods $V_1 \cap U$ and $V_2 \cap V$ of $p(x)$ and $p(y)$ which are evenly covered by p so that $p^{-1}(V_1 \cap U) = \sqcup_{\alpha_i} E_{\alpha_i}$ and $p^{-1}(V_2 \cap V) = \sqcup_{\alpha_j} F_{\alpha_j}$. The E_{α_i} containing x does not intersect the F_{α_j} containing y. Thus, E is Hausdorff.

(c) Suppose B is compact and all fibers of p are finite. Consider a covering $\mathcal{U} = \{U_\alpha\}_{\alpha\in A}$ of E. For each b, there is a finite subcover $\{U_{b_i}\}_{i=1_b^n} \subseteq \mathcal{U}$ of the fiber $p^{-1}(b)$. Let V_b be an open neighborhood of b in B such that $p^{-1}(V_b) \subseteq \cup_{i=1}^{n_n} U_{b_i}$. Since $\{V_b\}_{b\in B}$ covers B, the compactness of B yields a finite subcover $\{V_{b_k}\}_{k=1,\ldots,m}$ of B and its pre-image forms a cover of E, as well as $\{V_{b_k}\}_{k=1,\ldots,m}$. This yields a finite subcover in $\mathcal{U}$ of E.
Suppose instead that E is compact. Consider a cover $\mathcal{B} = \{B_\alpha\}_{\alpha\in A}$ of B. There exists a finite subcover $\mathcal{U} = \{U_i\}_{i=1}^n$ of $p^{-1}(B) = \{p^{-1}(B_\alpha)\}_{\alpha\in A}$. Since p is a surjective and open map, the images $p(\mathcal{U}) = \{p(U_i)\}_{i=1}^n$ form a finite subcover of B.
Each fiber has a finite subcover and discreteness implies that it is finite.

(d) Suppose E is Hausdorff and compact, and consider distinct points $b, c \in B$. By well-posedness and surjectivity of p, there exist distinct points $x, y \in E$ such that $p(x) = b$ and $p(y) = c$. By the Hausdorff axiom, there exist disjoint open neighborhoods U of x and V of y. They live in disjoint compact subspaces. By compactness, $p(U)$ and $p(V)$ are disjoint open sets containing $p(x)$ and $p(y)$.

□

3. (10 pts.) There are analogs for homology of many results for the fundamental group.

Theorem 3. *Let A be a subset of X, and let $i : A \to X$ be the inclusion map.*

(a) If $r : X \to A$ is a retraction, then the maps induced on all homology groups by the inclusion and retraction are injections and surjections, respectively.

(b) If A is a deformation retract of X, then the maps induced on all homology groups by the inclusion and deformation retraction are isomorphisms.

Proof. For some n, let $i_\star : H_n(A) \to H_n(X)$ be the map induced by inclusion and let $r_\star : H_n(X) \to H_n(A)$ be the map induced by a retraction $r : X \to A$.

(a) Let $z \in \ker(i_\star)$. Then, $(r_\star \circ i_\star) = r_\star(i_\star(z)) = i_\star(x) = 0$, so $z \in \ker(r_\star \circ i_\star)$ implies the following containment of groups $\ker(i_\star) \subseteq \ker(r_\star \circ i_\star) = 1_{H_n(A)} = \{0\}$. Since $\ker(i_\star) = 0$, the map $i_\star$ is injective.

Let z be in $H_n(A)$. Then, it is a cycle in $Z_n(A) \subseteq Z_n(X)$ with a boundary in $B_n(A) \subseteq B_n(X)$. Therefore, z is in $H_n(X)$. Thus, $r_\star$ is surjective.

(b) Suppose there is a deformation retraction $H : X \times I \to X$ of X onto A, and let $r = H(x, 1) : X \to A$. It suffices to show that $i_\star$ is surjective and that r is injective.

Notice that r induces a chain homomorphism $r_\# : C_n(X) \to C_n(A)$ by $r(\sigma) = r \circ \sigma$. This induces a homomorphism $r_\star : H_n(X) \to H_n(A)$ by $[r(a)] = [r_\#(a)]$.

According to Hatcher Theorem 2.10, the homotopic maps $[r \circ i] = 1$ induce the same homomorphism. We obtain $(i \circ r)_\star = 1_{X_\star}$ and thus $i_\star \circ r_\star = 1_{H_n(X)}$. Therefore, $i_\star$ is surjective and $r_\star$ is surjective.

□

4. (30 pts.) In this problem, we will take the first steps to understanding the higher homotopy groups, $\pi_n(X, x_0)$, for $n \geq 2$. Here $x_0 \in X$ is a fixed base point.

Let $I^n = I \times \cdots \times I$ be the n-fold product of intervals $I = [0, 1]$, and let ∂I^n denote its boundary in $\mathbb{R}^n$. We say f is a map of $(I^n, \partial I^n)$ into (X, x_0), and write

$$f : (I^n, \partial I^n) \to (X, x_0),$$

if $f : I^n \to X$ and $f(\partial I^n) = \{x_0\}$. We say two continuous maps $f, g : (I^n, \partial I^n) \to (X, x_0)$ are homotopic if there exists a homotopy

$$H : I^n \times I \to X$$

of f to g for which $H(x, t) = x_0$ if $x \in \partial I^n$. (More formally, we should say the maps are *homotopic relative to* ∂I^n to indicate the homotopy fixes ∂I^n, but for simplicity we drop the extra phrase here.)

It's straightforward to show that this notion of homotopic defines an equivalence relation, so let $[f]$ denote the equivalence class of such an f.

Theorem 4. *Let $\pi_n(X, x_0)$ be the collection of homotopy equivalence classes, which is, so far, only a set, and not yet a group. The following statements (a–e) are true.*

(a) If X is a convex subset of $\mathbb{R}^n$, then $\pi_n(X, x_0)$ has a single element.

(b) Define an operation $*$ on continuous maps of $(I^n, \partial I^n)$ into (X, x_0), by

$$f * g(t_1, t_2, \ldots, t_n) = \begin{cases} f(t_1, t_2, \ldots, t_{n-1}, 2t_n) & \text{if } t_n \leq 1/2, \\ g(t_1, t_2, \ldots, t_{n-1}, 2t_n - 1) & \text{if } t_n \geq 1/2. \end{cases}$$

The product $f * g$ is a continuous map of $(I^n, \partial I^n)$ into (X, x_0), and $*$ induces a well-defined operation on homotopy classes by $[f] * [g] = [f * g]$.

(c) The class $[e_{x_0}]$ of the constant map to x_0 is a left identity element for $*$ on homotopy classes. (It's also a right identity, but we will not prove that.)

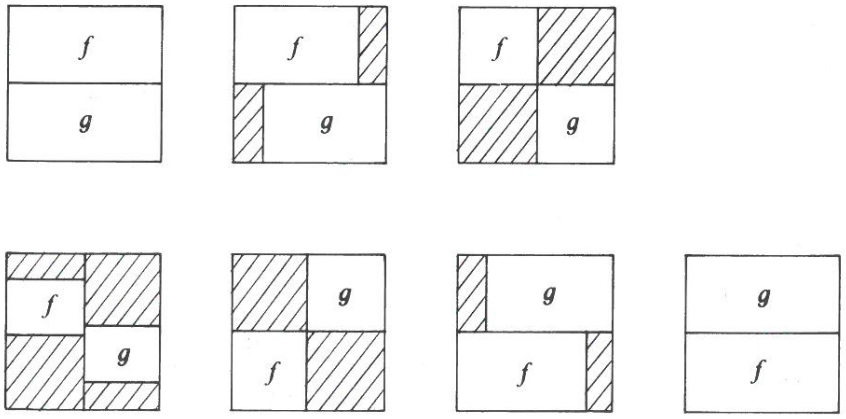
(d) For a continuous map $f : (I^n, \partial I^n) \to (X, x_0)$, continuous function $g : (I^n, \partial I^n) \to (X, x_0)$ given by the formula

$$g(t_1, t_2, \ldots, t_n) = f(t_1, t_2, \ldots, t_{n-1}, 1 - t_n)$$

satisfies $[f] * [g] = [e_{x_0}] = [g] * [f]$. Give a proof that $[f] * [g] = [e_{x_0}]$. (We will not prove $[g] * [f] = [e_{x_0}]$.)

Assuming $*$ is associative on homotopy classes (which I will NOT ask you to prove), the last 3 subproblems show $\pi_n(X, x_0)$ is a group.

(e) For $n > 1$, $\pi_n(X, x_0)$ is always abelian. The following pictures hint at how this can be proved for the $n = 2$ case. To show you understand, give a formula for a homotopy that does what is depicted in transitioning across the top 3 pictures *only.*



Proof. Let X be a space.

(a) Suppose X is a convex subset of $\mathbb{R}^n$. For any two continuous maps

$$f, g : (I^n, \partial I^n) \to (X, x_0),$$

we construct a homotopy $H : I^n \times I \to X$ from f to g for which $H(x,t) = x_0$ if $x \in \partial I^n$.

In particular, consider the straight-line map given by $H(s,t) = (1-t)f(s) + tg(s)$ for $s \in I^n$ and $t \in I$. Convexity of X implies that H is a homotopy. If $x \in \partial I^n$, then $H(x,t) = (1-t)f(x) + tg(x) = (1-t)x_0 + tx_0 = (1-t+t)x_0 = x_0$. Since any two continuous maps are homotopic, it therefore follows that the homotopy group $\pi_n(X, x_0)$ has a single element.

(b) We show the product $f * g$ is a continuous map of $(I^n, \partial I^n)$ into (X, x_0) by applying the closed pasting lemma, and then show the operation $*$ is well defined on homotopy classes by constructing the required homotopy.

Notice that the product is a continuous function defined on two closed sets: $A = I^{n-1} \times \left[0, \frac{1}{2}\right]$ and $B = I^{n-1} \times \left[\frac{1}{2}, 1\right]$. On each closed set, its continuity follows from the fact that it is equal to a composition of continuous functions, so the operation $*$ is continuous on I^n by the closed pasting lemma. If $x \in \partial I^n$, then $(f * g)(x) \in \{f(x), g(x)\} = \{0\}$, so $f * g$ is a continuous map of $(I^n, \partial I^n)$ into (X, x_0).

Next we show that $[f] = [h]$ and $[g] = [k]$ together imply that $[h] * [k] = [h * k] = [f * g] = [f] * [g]$. In particular, suppose there are homotopies H and K from f to h, and from g to k. Then, there is a homotopy $(H * K) : I^n \times I \to X$

$$(H * K)(t_1, \ldots, t_n, t) = \begin{cases} H(t_1, t_2, \ldots, t_{n-1}, 2t_n, t) & \text{if } t_n \leq 1/2, \\ K(t_1, t_2, \ldots, t_{n-1}, 2t_n - 1, t) & \text{if } t_n \geq 1/2. \end{cases}$$

which achieves the following values:

$$\begin{aligned} (H * K)(\cdot, 0, t) &= H(\cdot, 0, t) = f(\cdot, 0) = (f * g)(cdot, 0) \\ (H * K)(\cdot, 1/2, t) &= H(\cdot, 1/2, t) = g(\cdot, 1) = h(\cdot, 0) = K(\cdot, 1/2, t) \\ (H * K)(\cdot, 1, t) &= K(\cdot, 1, t) = k(\cdot, 1) = (h * k)(\cdot, 1), \end{aligned}$$

making it a homotopy from $f * g$ to $h * k$.

(c) Then,

$$[e_{x_0} * f](t_1, t_2, \ldots, t_n) = \begin{cases} [e_{x_0}](t_1, t_2, \ldots, t_{n-1}, 2t_n) & \text{if } t_n \leq 1/2, \\ [f](t_1, t_2, \ldots, t_{n-1}, 2t_n - 1) & \text{if } t_n \geq 1/2. \end{cases}$$
$$= \begin{cases} x_0 & \text{if } t_n \leq 1/2, \\ [f](t_1, t_2, \ldots, t_{n-1}, 2t_n - 1) & \text{if } t_n \geq 1/2. \end{cases}$$
$$= [f].$$

(d) Notice that

$$[f * g](t_1, t_2, \ldots, t_n) = \begin{cases} [f](t_1, t_2, \ldots, t_{n-1}, 2t_n) & \text{if } t_n \leq 1/2, \\ [g](t_1, t_2, \ldots, t_{n-1}, 2t_n - 1) & \text{if } t_n \geq 1/2. \end{cases}$$
$$= \begin{cases} [f](t_1, t_2, \ldots, t_{n-1}, 2t_n) & \text{if } t_n \leq 1/2, \\ [f](t_1, t_2, \ldots, t_{n-1}, 1 - (2t_n - 1))) & \text{if } t_n \geq 1/2. \end{cases}.$$

The first $(n-1)$ coordinates are contractible, and the nth forms a contractible loop as well.

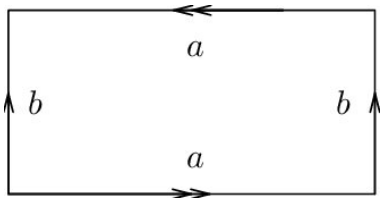
(e) We construct a homotopy $H : I^2 \times I \to X$ where $H(t_1, t_2, 0)$ denotes the first square and $H(t_1, t_2, 1)$ denotes the third square, and t_3 moves through the page. Let t_1 be the coordinate which traverses a given cross-section from left to right and t_2 be the coordinate which traverses a given cross-section from bottom to top.

Then, the homotopy is given below.

$$\begin{cases} \begin{cases} x_0 & \text{for } t_1 \leq t_3 \\ g((1 - e^(t_1 - t_3))/(t_1 - e^(1 - t_3)), 2t_2) & \text{for } t_3 \leq t_1 \leq 1 \end{cases} & \text{for } t_2 \leq 1/2 \\ \begin{cases} f(t_1/t_3, 2t_2 - 1) & \text{for } t_1 \leq t_3 \\ x_0 & \text{for } t_1 \geq t_3 \end{cases} & \text{for } t_2 \geq 1/2 \end{cases}$$

□

5. (30 pts.) Let X be the *Klein bottle*, a 2-dimensional surface obtained as a quotient of a rectangle $R = A \times B$ by identifying edges as shown in the figure below. Let a and b be loops in X based at x_0, with images as shown in the figure, so $[a], [b] \in \pi_1(X, x_0)$.



Theorem 5. *The following are true:*

(a) Explain informally why X has $\mathbb{R}^2$ as its universal cover, including giving a verbal description of the covering map (no formula, or formal proof that is a covering map need be given). Give enough examples of evenly covered subsets of X (depicted as subsets of the rectangle with identifications) and associated slices to convince a reader you understand the covering map well. Point out any significant ways in which this covering differs from the covering of the torus T by $\mathbb{R}^2$.

(b) Explain informally how X has T as a 2-fold cover. Explain how this can be seen through the universal covers of X and T.

(c) In $\pi_1(X, x_0)$ neither $[a]$ nor $[b]$ is the identity, nor equal to their own inverses, and that $[a] * [b] = [b] * [a]^{-1}$.

(d) The elements of $\pi_1(X, x_0)$ can be identified with elements of $\mathbb{Z}^2$ as a set*, but the correspondence does* not *yield a group isomomorphism.*

(e) Using this identification with $\mathbb{Z}^2$, the group operation in $\pi_1(X, x_0)$ is expressed as

$$(n, m)(k, \ell) = (n + \phi_m(k), m + \ell),$$

where $\phi_m : \mathbb{Z} \to \mathbb{Z}$ is the identity map for m even, and the inverse map for m odd.

(f) For any two groups G,H, with $\mathrm{Aut}(G)$ denoting the group of automorphisms of G, let

$$\phi : H \to \mathrm{Aut}(G)$$

be any homomorphism. Then the semidirect product $G \rtimes_\phi H$ with respect to ϕ *is defined as the set $G \times H$ together with the operation*

$$(g_1, h_1)(g_2, h_2) = (g_1 \phi(h_1)(g_2), h_1 h_2).$$

The semidirect product $\mathbb{Z} \rtimes_\phi \mathbb{Z}$ is defined for two ϕ maps, and one of them is the fundamental group of X.

Proof. Let X be the Klein bottle and recall that the plane $\mathbb{R}^2$ is simply connected.

(a) We explain informally why X has $\mathbb{R}^2$ as its universal cover by showing how to tile the plane with the rectangle R as in Figure 1

We explain why X has $\mathbb{R}^2$ as its universal cover by recalling that $\mathbb{R}^2$ is simply connected and verbally describing the associated covering map $p : \mathbb{R}^2 \to X$.

Without loss of generality, we assume $x_0 = (0, 0)$ and that the rectangle is given by $R = A \times B$, where $A = [0, h]$ and $B = [0, v]$. Let $e_0 = (0, 0)$ and $p(e_0) = x_0$. Let $p(e) = q(e)$, where q is the quotient map of the rectangle R and e is in R. We tile the band $B_1 = \mathbb{R} \times B \subseteq \mathbb{R}^2$ by repeating copies of R, shifted horizontally by $|A|$. We tile the band $B_2 = \mathbb{R} \times [v, 2v]$ by reflecting band B_1.

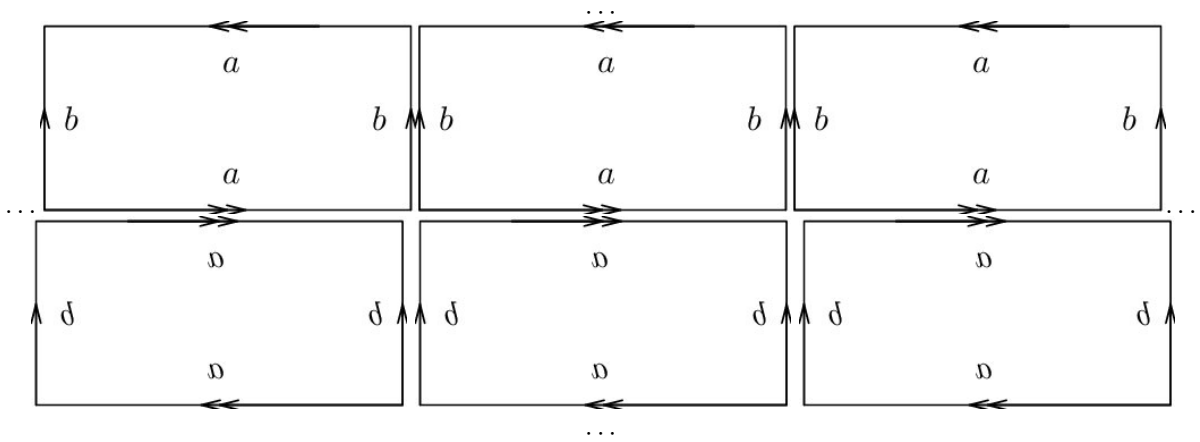


Figure 1: Caption

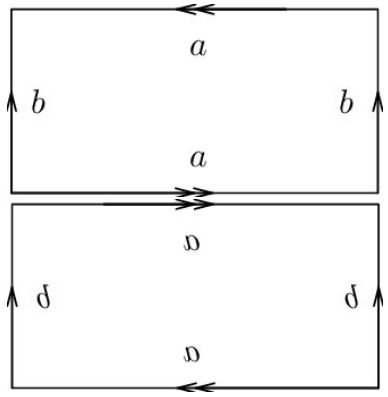


Figure 2: Caption

Since each point e in the plane is in some rectangle, each point has a representative $[e]$ in R, so we assign $p(e) = p([e])$.

The basepoint x_0, for example, is covered by $(0, 0)$ as well as (e_1, e_2) for multiples $e_1 = nh$ and $e_2 = mh$ for all integers $n, m \in \mathbb{Z}$. The point in the center of the rectangle $(h/2, v/2)$ is covered by points of the form $(nh + h/2, mh + v/2)$ for all integers $n, m \in \mathbb{Z}$. Any point in the Klein bottle has an associated point on the rectangle, and its fiber is given by the coset of the fiber of the basepoint added corresponding to the addition of the point on the rectangle.

(b) Consider a horizontal reflection of the rectangle stacked on top of the rectangle, as in Figure 2, with the overlapping edges a identified (which necessarily identifies the four outer corners). This shows how to see the torus as a copy of two klein bottles. Alternatively, the covering space of the torus can be seen as the covering space of the klein bottle with vertical units multiplied by $2v$ instead of v.

(c) Following the diagram, it is clear that $[a] * [b] * [a]^{-1} = [b]$ as there is a homotopy which slowly contracts $[a] * [b] * [a]^{-1} * [b]^{-1}$ to the constant loop $[e_{x_0}]$.

By the Seifert-van Kampen theorem, the fundamental group of X is isomorphic to $\mathbb{Z}^2/\langle abab^{-1}\rangle$, which notably lacks relations which restrict the order of powers of a or powers of b.

Therefore, they each have infinite order. If either were equal to the identity or their own inverses,

they would have order 1 or 2, which would be contradictions.

(d) A loop in X can be lifted to a path in the universal cover which starts at $(0,0)$ and ends at (nu, mv) for unique integers $n, m \in \mathbb{Z}$, and thus elements of the fundamental group can be identified with elements (n,m) of the set $\mathbb{Z}^2$. However, $\mathbb{Z}^2$ is an abelian group, while the result in part (c) show that $[a]$ and $[b]$ fail to commute:

$$[a] * [b] = [b] * [a]^{-1} \neq [b] * [a].$$

Since commutativity is a property preserved by homomorphisms, it follows that the set correspondence above is not a group isomorphism.

(e) The multiplication $([a]^n * [b]^m)([a]^k * [b]^l)$ corresponds to starting at $[a]^n * [b]^m$, moving l units in the direction of $[b]$ which is positively upward and moving k units in the direction of $[a]$ which is positively right-ward if m is even and positively left-ward (negatively right-ward) if m is odd.

This is precisely captured by the formula $(n,m)(k,l) = (n + \phi_m(k), m + l)$, where $\phi_m : \mathbb{Z} \to \mathbb{Z}$ is the identity map for even m and the inverse map for odd m.

(f) We show (i) that there are two maps $\phi_{\pm 1} : \mathbb{Z} \to \mathrm{Aut}(\mathbb{Z})$ with respect to which the semidirect product $\mathbb{Z} \rtimes_\phi \mathbb{Z}$ is defined, and (ii) conclude that each of them yield the fundamental group of a known space.

i. First, we show there are two automorphisms $\psi_{\pm 1} : \mathbb{Z} \to \mathbb{Z}$ in $\mathrm{Aut}(\mathbb{Z})$ by recalling that an arbitrary automorphism $\psi : Z \to \mathbb{Z}$ is uniquely determined by the value assigned to a generator, say $\psi(1)$ without loss of generality. Surjectivity demands $\psi(\mathbb{Z}) = \psi(1)\mathbb{Z} = \mathbb{Z}$ and thus that $\psi_{\pm 1}(1) = \pm 1$ which extend to $\psi_{\pm 1}(n) = \pm n$.
Next we show there are two homomorphisms $\phi_{\pm 1} : \mathbb{Z} \to \mathrm{Aut}(\mathbb{Z})$, again by recalling that an arbitrary homomorphism $\phi : \mathbb{Z} \to \mathrm{Aut}(\mathbb{Z})$ is uniquely determined by the value assigned to a generator, say $\phi(1)$ without loss of generality. Without loss of generality, we obtain $\phi_{\pm 1}(1) = \psi_{\pm 1}$. Clearly, $\phi_1(h_1)(g_2) = \psi_1(g_2) = g_2$. By linearity, we obtain $\phi_{-1}(h_1)(g_2) = \psi_{(-1)^{h_1}}(g_2) = (-1)^{h_1} g_2$.
The semidirect product $\mathbb{Z} \rtimes_{\phi_1} \mathbb{Z}$ with respect to ϕ_1 is equipped with the following operation

$$(g_1, h_1)(g_2, h_2) = (g_1 + \phi_1(h_1)(g_2), h_1 + h_2) = (g_1 + \psi_1(g_2), h_1 + h_2) = (g_1 + g_2, h_1 + h_2),$$

which is precisely the operation on the fundamental group $\mathbb{Z} \times \mathbb{Z}$ of the torus.

ii. Let $\phi = \phi_{-1}$. Then, the semidirect product $\mathbb{Z} \rtimes_\phi \mathbb{Z}$ with respect to ϕ is equipped with the following operation

$$(g_1, h_1)(g_2, h_2) = (g_1 + \phi(h_1)(g_2), h_1 + h_2) = (g_1 + \phi_{h_1}(g_2), h_1 + h_2),$$

where $\phi_{h_1}(g_2) = \phi(h_1)(g_2) = \psi_{(-1)^{h_1}}(g_2) = (-1)^{h_1} g_2$ is the identity map for even h_1 and the inverse map for odd h_1.

Up to relabeling, the semidirect product $\mathbb{Z} \rtimes_{\phi_{-1}} \mathbb{Z}$ with respect to ϕ_{-1} is precisely the group operation in $\pi_1(X, x_0)$.

Therefore, the fundamental group of the Klein bottle is the uniquely non-trivial semidirect product of $\mathbb{Z}$ with itself. □